

# Jackson Bopp

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Computer Engineering student seeking an entry-level embedded systems or FPGA/hardware engineering role, with hands-on experience in digital design, firmware, and hardware debugging.

## Skills

**Languages:** C, C++, Python, Verilog/VHDL, RISC-V Assembly, MATLAB, Bash

**Hardware:** MSP430FR2355, Raspberry Pi, Arduino, FPGA (Xilinx), L298N, HM-10 BLE

**Protocols:** UART, I2C, SPI, PWM, MQTT, GPIO, Bluetooth LE

**Tools:** Git, Docker, Linux, Cursor, GDB, Xilinx ISE, SOLIDWORKS

**Coursework:** Embedded Systems, Computer Architecture, CMOS-VLSI Design, FPGA Design, Operating Systems, Computer System Design, Trustworthy Infrastructures

**Certifications:** Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

## Education

**University of South Florida**

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52/4.00

Tampa, FL

Graduating Fall 2026

## Projects

### FPGA Audio Message Recorder

Spring 2026

*Xilinx Spartan-6, Verilog, PicoBlaze, DDR2 SDRAM, I2C, UART*

- Collaborated with a 3-member team to design, synthesize, and deploy a full audio recording and playback system on a Xilinx Spartan-6 FPGA.
- Wrote the top-level Verilog FSM managing record, play, pause, rewind, and a 4-message library with independent delete, backed by a DDR2 SDRAM interface for sample storage and an I2C master driving the SSM2603 audio codec.
- Programmed a PicoBlaze embedded soft-core in assembly for the UART menu, implemented hardware volume scaling, and debugged real hardware issues (audio distortion, a rewind pointer bug) across several controller revisions to reach a stable final build.

### Testbench

July 2026

*MSP430FR2355, ESP32, STM32F103C8, RP2040, MQTT, Raspberry Pi, Docker*

- Designed a hardware-agnostic test platform that streams live telemetry from UART-based microcontrollers through a Raspberry Pi bridge over MQTT to a web dashboard.
- Implemented reference firmware for the MSP430FR2355, ESP32, STM32F103C8, and RP2040, each emitting the same JSON-over-UART frame format so any of the four boards works with the same bridge and backend.
- Developed the Python MQTT bridge and a hardware simulator, plus a capture-and-replay tool for running demos and regression tests without physical hardware.
- Added schema-validated frame parsing and a cross-platform test suite checking that all four firmware builds produce frames the backend accepts.
- Set up a GitHub Actions workflow that spins up Postgres and Mosquitto containers and runs an end-to-end simulated hardware test on every push.

## Open Source

### Open Source Contributions

August 2026

*C++, Go, Python, D-Bus, Linux, Gerrit and GitHub code review*

- **OpenBMC phosphor-user-manager:** diagnosed and fixed two bugs in a BMC user management daemon, an unhandled exception that could abort account setup for every user over one bad record, and a C++ dispatch issue causing tests to silently run against the real system instead of mocks; submitted with full before and after test verification.
- **Zephyr RTOS:** found and fixed a devicetree macro bug in the MIPI DBI display driver that produced an incorrect chip select GPIO check for out of range devices; submitted with a compile time regression test.
- **TinyGo:** found an unsynchronized write in the runtime's timer reset path racing against every other locked access to the same data; built the compiler from source to verify, and submitted the fix.
- **PlatformIO:** root-caused and fixed a crash in the serial device monitor triggered by non-interactive input; merged upstream with an added regression test.

## Experience

### City of Winter Haven Technology Services

April 2026 – Present

*Smart City Student Volunteer*

Winter Haven, FL

- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Configured and optimized PyTorch and TensorFlow environments to support the city's early-stage AI hosting infrastructure, and resolved day-to-day IT support tickets.

### Subs N' Such

April 2023 – March 2026

*Manager, Shift Leader, and Front-End Web Support*

Lutz, FL

- Built the front-end of the restaurant's online ordering site within Square's third-party UI framework, working around its layout and customization constraints to deliver a functional ordering experience.
- Managed shift operations and enforced food safety procedures.

## Involvement

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE